Jenkins-02 — Jenkins Administration & Backup Documentation

1. Objective

The objective of this lab is to configure Jenkins agents, integrate a Jenkins job with GitHub webhooks, configure automated job triggers, back up the Jenkins server using two different methods, and restore the Jenkins backup on a newly created Jenkins server.

Task 1 — Configure 2 Slave Machines

Objective

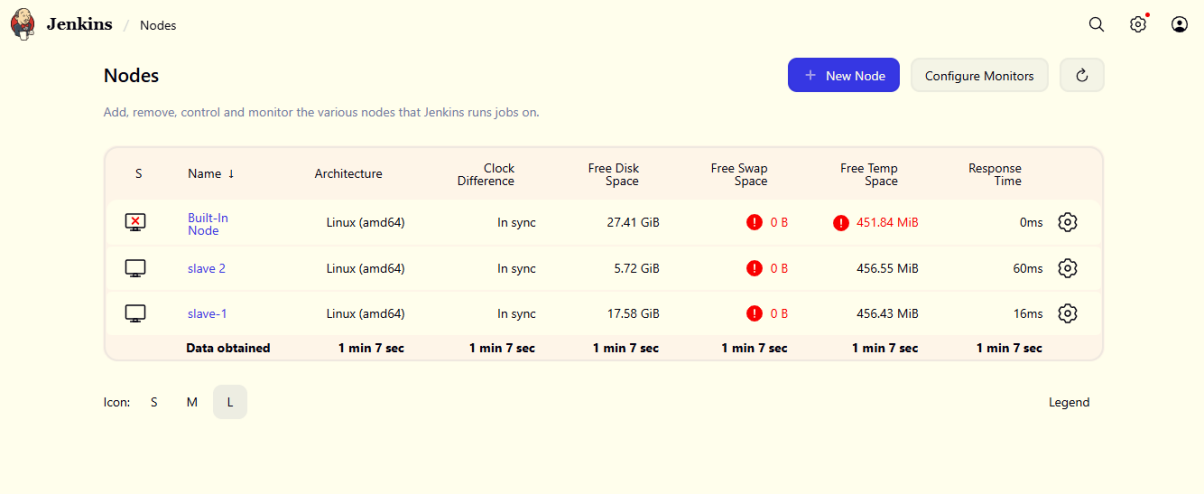
Configure two EC2 machines as Jenkins agents/slaves and connect them to the Jenkins master using SSH.

Steps Performed

Agent-1

Java was installed on the first agent:





The SSH credentials for the respective EC2 instances were configured.

Result

Both Jenkins agents successfully connected to the Jenkins master.

Task 2 — Configure Webhook for Jenkins Job

Objective

Configure a GitHub webhook so that a push to the GitHub repository can trigger the Jenkins job.

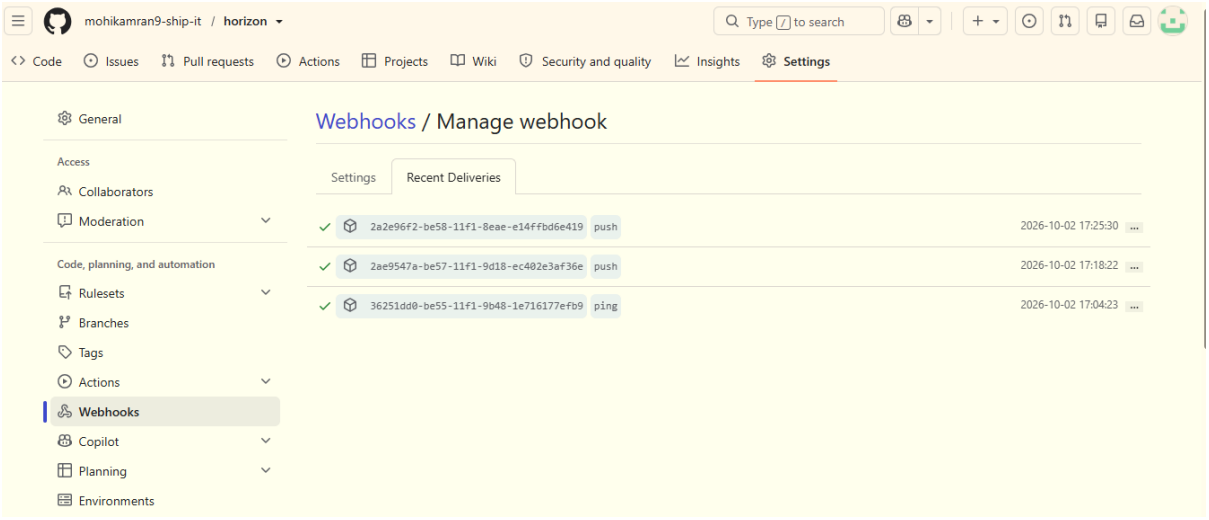
Jenkins Configuration

The Jenkins job was opened and configured with the GitHub repository.

Under **Build Triggers**, the GitHub webhook trigger was enabled.

Example:

GitHub hook trigger for GITScm polling



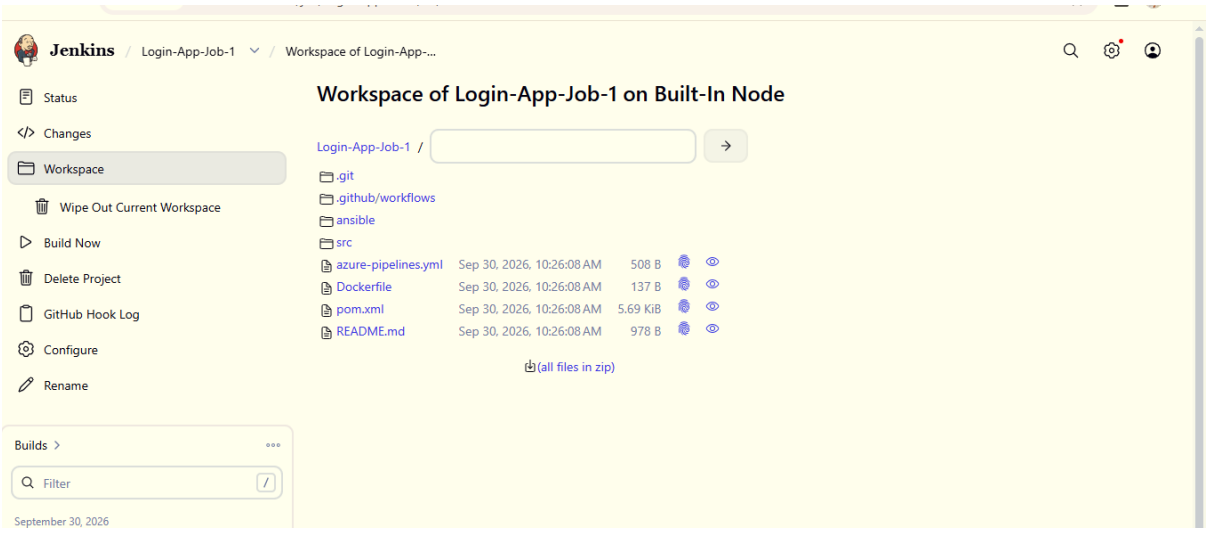
Testing

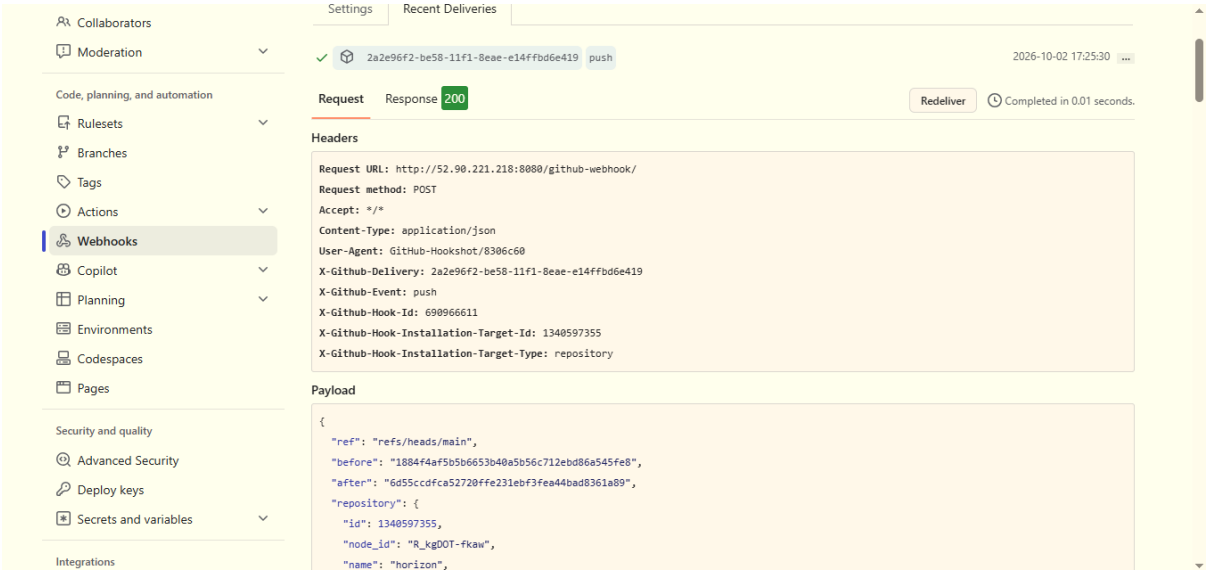
A change was committed and pushed to the GitHub repository.

The GitHub webhook sent the event to Jenkins and the Jenkins job was triggered.

Result

GitHub webhook integration with Jenkins was successfully configured and tested.





Task 4 — Jenkins Backup Using Bash Script

Objective

Create a Bash script to back up Jenkins configuration, jobs, credentials/configuration files, plugins, and other Jenkins data.

The Jenkins home directory used in this lab was:

```
[root@ip-172-31-27-94 jenkins]# sudo mkdir -p /opt/jenkins-backups
[root@ip-172-31-27-94 jenkins]# sudo nano /usr/local/bin/jenkins-backup.sh
[root@ip-172-31-27-94 jenkins]# sudo chmod +x /usr/local/bin/jenkins-backup.sh
[root@ip-172-31-27-94 jenkins]# sudo /usr/local/bin/jenkins-backup.sh
Starting Jenkins backup...
tar: Removing leading `/' from member names
Backup completed:
/opt/jenkins-backups/jenkins-backup-2026-10-02_12-27-59.tar.gz
[root@ip-172-31-27-94 jenkins]# ls -lh
total 104K
-rw-r--r--.  1 jenkins jenkins 1.6K Oct  2 10:48 config.xml
-rw-r--r--.  1 jenkins jenkins 4.1K Oct  1 05:58 credentials.xml
drwxr-xr-x.  5 jenkins jenkins   36 Oct  1 06:01 fingerprints
-rw-r--r--.  1 jenkins jenkins  156 Oct  2 10:48 hudson.model.UpdateCenter.xml
-rw-r--r--.  1 jenkins jenkins  370 Sep 30 07:28 hudson.plugins.git.GitTool.xml
-rw-------.  1 jenkins jenkins 1.7K Sep 30 07:28 identity.key.enc
-rw-r--r--.  1 jenkins jenkins    7 Oct  2 10:48 jenkins.install.InstallUtil.lastExecVersion
-rw-r--r--.  1 jenkins jenkins    7 Sep 30 07:29 jenkins.install.UpgradeWizard.state
-rw-r--r--.  1 jenkins jenkins  182 Sep 30 07:29 jenkins.model.JenkinsLocationConfiguration.xml
-rw-r--r--.  1 jenkins jenkins  171 Sep 30 07:26 jenkins.telemetry.Correlator.xml
drwxr-xr-x.  5 jenkins jenkins   75 Sep 30 12:22 jobs
drwxr-xr-x.  3 jenkins jenkins  110 Oct  2 11:48 logs
-rw-r--r--.  1 jenkins jenkins 1.1K Oct  2 10:48 nodeMonitors.xml
drwxr-xr-x.  4 jenkins jenkins   36 Oct  1 06:16 nodes
drwxr-xr-x. 94 jenkins jenkins  16K Sep 30 07:28 plugins
-rw-r--r--.  1 jenkins jenkins  976 Oct  2 12:16 queue.xml
-rw-r--r--.  1 jenkins jenkins  258 Oct  2 08:28 queue.xml.bak
-rw-r--r--.  1 jenkins jenkins   64 Sep 30 07:26 secret.key
-rw-r--r--.  1 jenkins jenkins    0 Sep 30 07:26 secret.key.not-so-secret
drwx------.  2 jenkins jenkins  16K Sep 30 12:38 secrets
drwxr-xr-x.  2 jenkins jenkins  149 Oct  2 06:15 updates
drwxr-xr-x.  2 jenkins jenkins   24 Sep 30 07:26 userContent
drwxr-xr-x.  5 jenkins jenkins  16K Sep 30 09:12 users
drwxr-xr-x.  5 jenkins jenkins   75 Sep 30 12:22 workspace
[root@ip-172-31-27-94 jenkins]#
```

Task 5 — Jenkins Backup Using ThinBackup Plugin

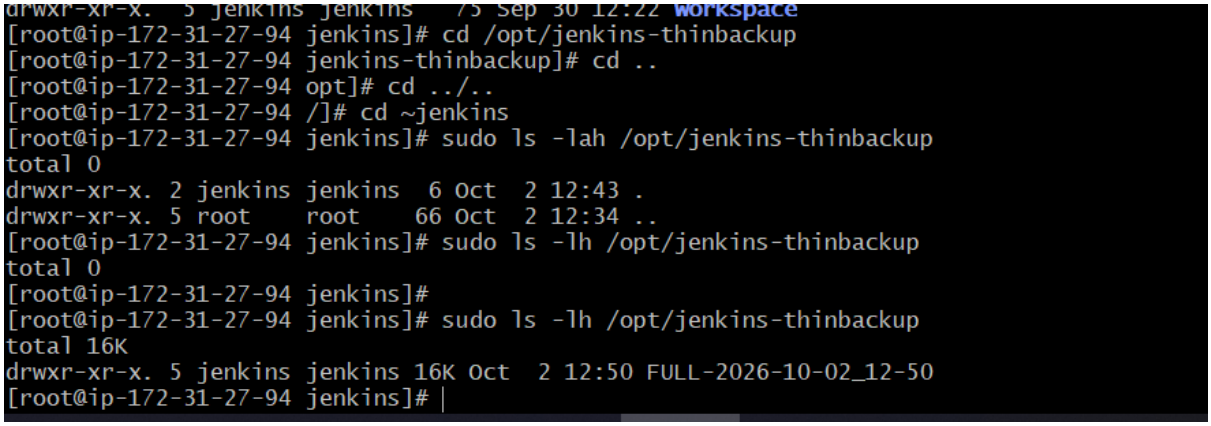
Objective

Configure the **ThinBackup** plugin to automatically back up Jenkins data.

Plugin Installation

Navigate to:

Manage Jenkins → Plugins → Available plugins



Backup Verification

The backup directory was checked:

```
ls -lh /opt/jenkins-thinbackup/
```

Result

ThinBackup was successfully installed and configured, and a Jenkins backup was created using the plugin.

Screenshots to add:

📸 ThinBackup plugin installation
📸 ThinBackup configuration page
📸 Backup directory
📸 Successful ThinBackup backup

Task 6 — Created New Jenkins Server

Objective

Create a new Jenkins server and restore the Jenkins configuration from the Bash-script backup.

A new EC2 instance was launched for the restore operation.

The required Java version was installed:

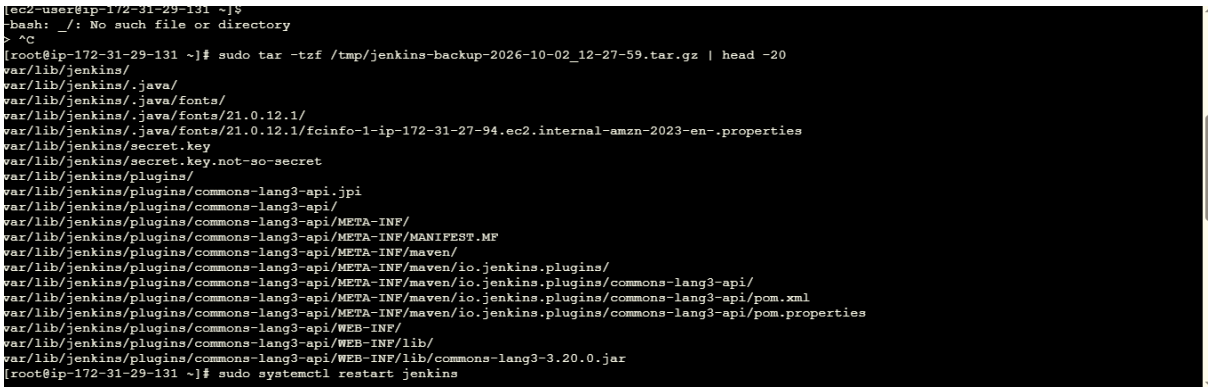
Jenkins was installed and started.

The Jenkins service was checked:

```
sudo systemctl status Jenkins
```

Transfer Jenkins Backup to New Server

The backup file created on the original Jenkins server was transferred to the new Jenkins server.



The Jenkins administration lab was successfully completed. Two Jenkins agents were connected to the master, GitHub webhook integration was configured, and automated job triggering was implemented using Poll SCM and Build Periodically. Jenkins data was backed up using both a Bash script and the ThinBackup plugin. Finally, a new Jenkins server was created and the Bash-script backup was restored to validate the backup and disaster-recovery process.